



Big Data Analytics

ML and Big Data Application

Contents

➤ ML

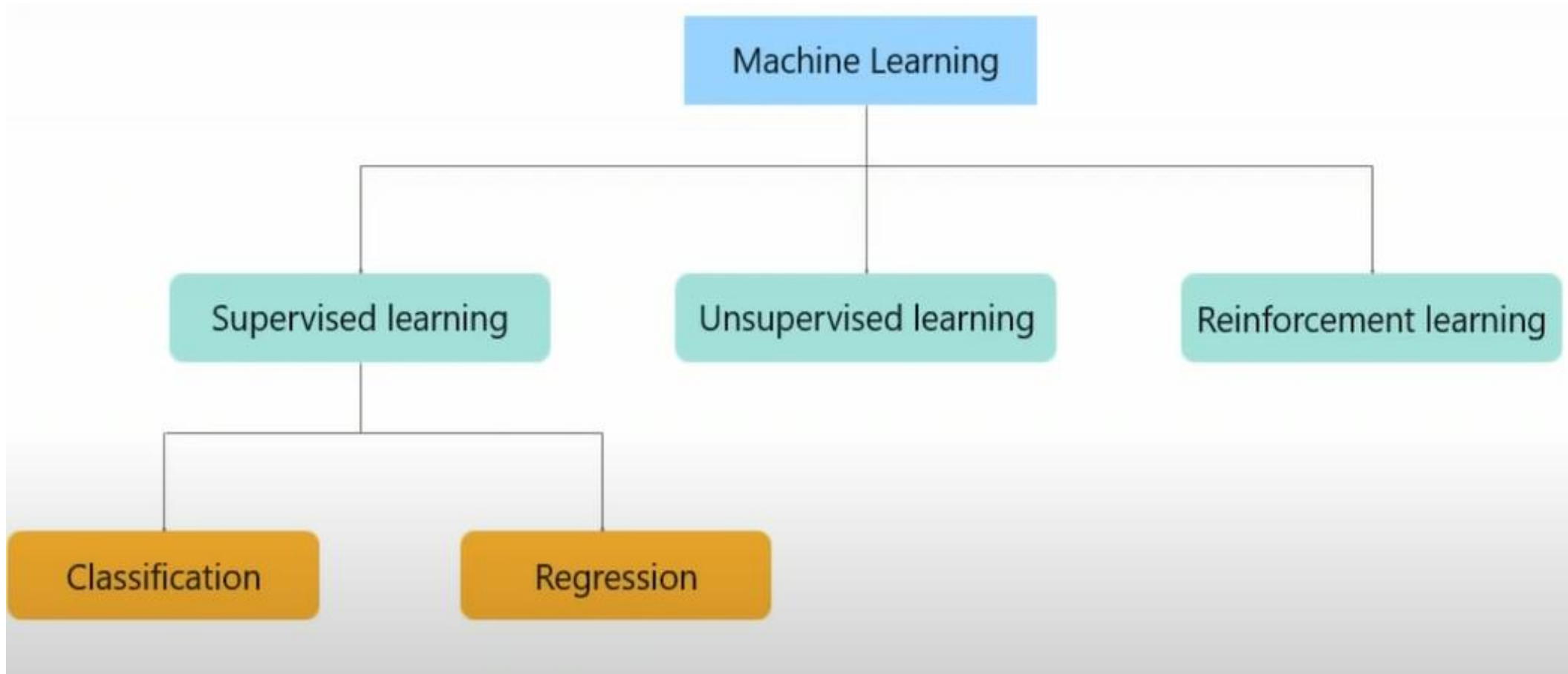
➤ Big Data Application

- Text Analytics
- Image Analytics
- Stream Analytics

Machine Learning

- Machine learning is the practice of programming computers to learn from data.
- In machine learning, data referred to as **training sets** or **examples**.
- When you have a problem that requires many long lists of rules to find the solution, In this case, machine-learning techniques can simplify your code and improve performance.

Machine Learning



Supervised Learning

- **Supervised Learning** is a machine learning technique that uses a collection of paired **input-output training** samples to learn about a system's input-output connection.
- Classification and regression are two types of problems that can be solved with supervised learning.
- **Supervision**: The training data (observations, measurements, etc.) are accompanied by **labels** indicating the class of the observations.
- New data is classified based on the training set.

Supervised Learning

- **Classification**

- Output type: discrete (binary/multi-classes)
- Trying to find: a boundary
- Evaluation: accuracy

- **Regression**

- Output type: continuous
- Trying to find: best fit line
- Evaluation: sum of squared errors



Regression

What is the temperature going to be tomorrow?

PREDICTION
84°



Classification

Will it be Cold or Hot tomorrow?

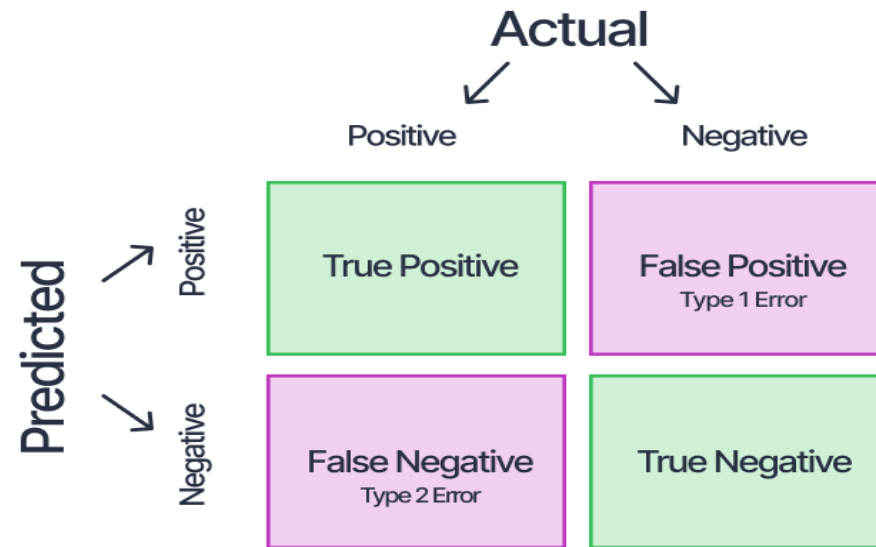
COLD

PREDICTION
HOT



Evaluation Matrices

Confusion Matrix



Cont..

- **TP:** True Positive: Predicted values correctly predicted as actual positive
- **FP:** Predicted values incorrectly predicted an actual positive. i.e., Negative values predicted as positive
- **FN:** False Negative: Positive values predicted as negative
- **TN:** True Negative: Predicted values correctly predicted as an actual negative.

Evaluation Matrices

<i>Measure</i>	<i>Formula</i>
accuracy, recognition rate	$\frac{TP + TN}{P + N}$
error rate, misclassification rate	$\frac{FP + FN}{P + N}$
sensitivity, true positive rate, recall	$\frac{TP}{P}$
specificity, true negative rate	$\frac{TN}{N}$
precision	$\frac{TP}{TP + FP}$
F , F_1 , F -score, harmonic mean of precision and recall	$\frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$
F_β , where β is a non-negative real number	$\frac{(1 + \beta^2) \times \text{precision} \times \text{recall}}{\beta^2 \times \text{precision} + \text{recall}}$

Example confusion matrix for a binary classifier

n=165	Predicted: NO	Predicted: YES
	Actual: NO	Actual: YES
	50	10
	5	100

What can we learn from this matrix?

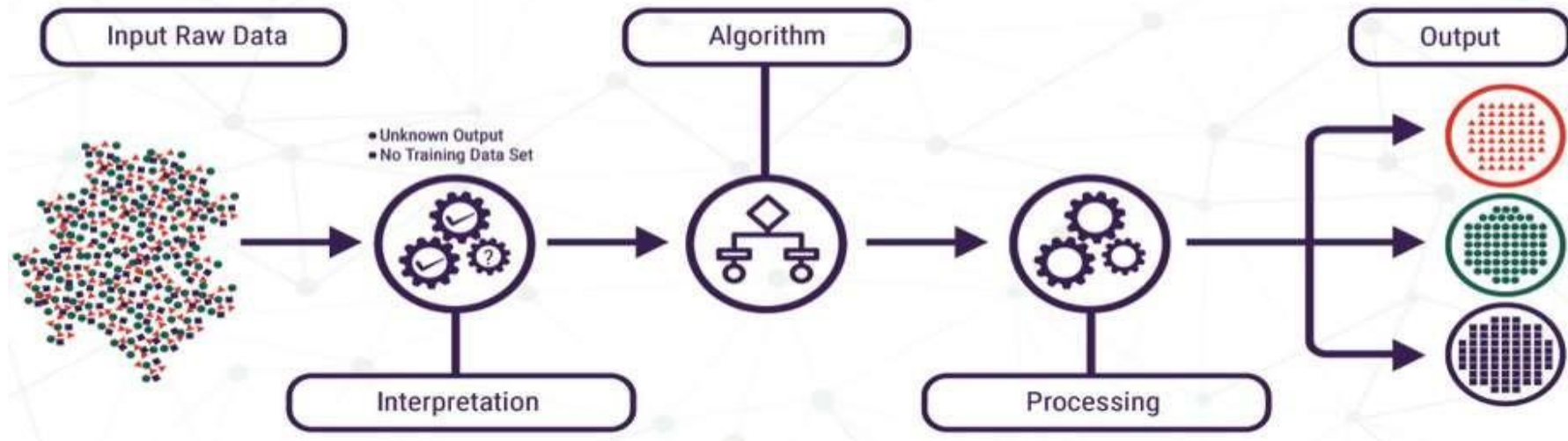
- There are two possible predicted classes: "yes" and "no". If we were predicting the presence of a disease, for example, "yes" would mean they have the disease, and "no" would mean they don't have the disease.
- The classifier made a total of 165 predictions (e.g., 165 patients were being tested for the presence of that disease).
- Out of those 165 cases, the classifier predicted "yes" 110 times, and "no" 55 times.
- In reality, 105 patients in the sample have the disease, and 60 patients do not.

n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

- **Accuracy:** Overall, how often is the classifier correct?
 - $(TP+TN)/total = (100+50)/165 = 0.91$
- **Misclassification Rate:** Overall, how often is it wrong?
 - $(FP+FN)/total = (10+5)/165 = 0.09$
 - equivalent to 1 minus Accuracy
 - also known as "Error Rate"
- **True Positive Rate:** When it's actually yes, how often does it predict yes?
 - $TP/actual\ yes = 100/105 = 0.95$
 - also known as "Sensitivity" or "Recall"
- **False Positive Rate:** When it's actually no, how often does it predict yes?
 - $FP/actual\ no = 10/60 = 0.17$
- **True Negative Rate:** When it's actually no, how often does it predict no?
 - $TN/actual\ no = 50/60 = 0.83$
 - equivalent to 1 minus False Positive Rate
 - also known as "Specificity"
- **Precision:** When it predicts yes, how often is it correct?
 - $TP/predicted\ yes = 100/110 = 0.91$

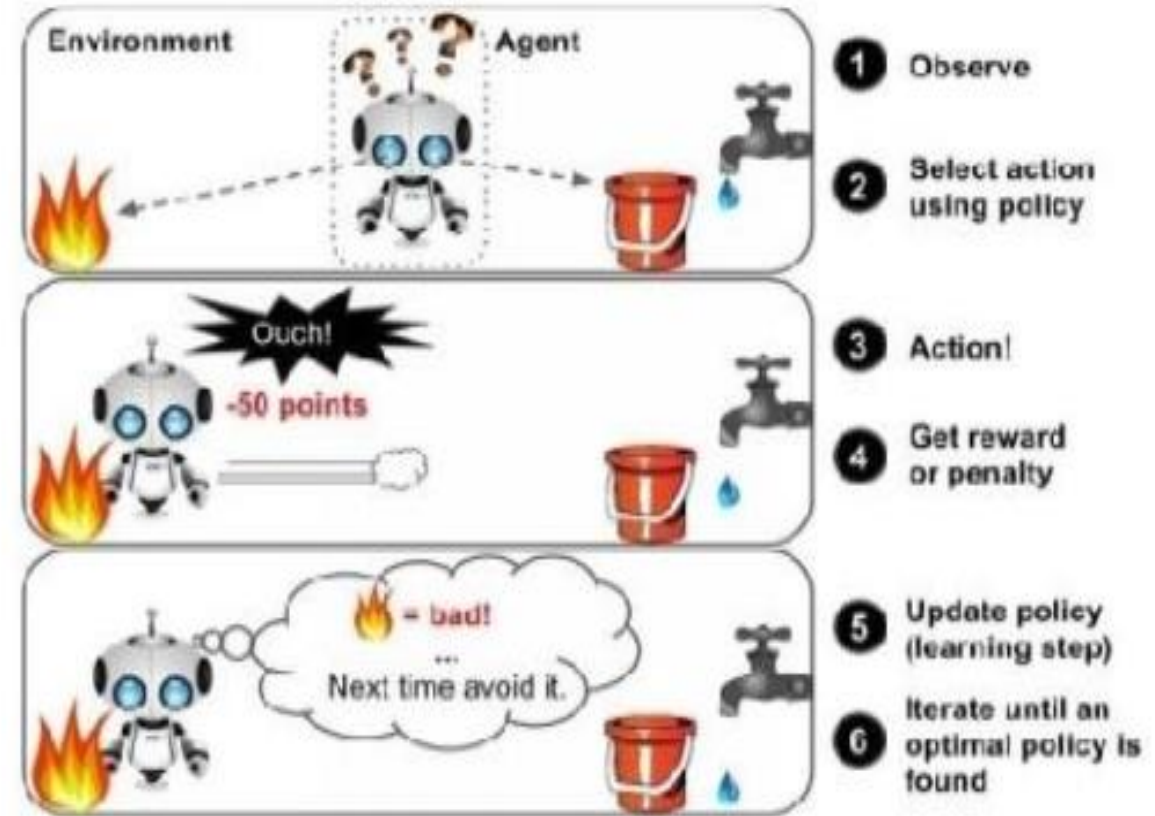
Unsupervised learning

- **No labels** are given to the learning algorithm.
- Find interesting information or pattern in the input data.
- Example: **clustering, dimensionality reduction, association rule mining**



Reinforcement learning

- A computer program interacts with a **dynamic environment** in which it must perform a certain action.
 - Learning what actions to take based on a reward signal
 - Involves an assumption of being in a state S_1 and evaluating the chance of changing to another S_2
 - In the end (along the way) a reward signal is received.
 - Want to take action to get the most reward possible.
 - The objective is to maximize the long-term cumulative reward.



Big Data Applications

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Applications of Text Analytics

- **Word frequency:** used to identify the most recurrent terms or concepts in a set of data.
 - ✓ Useful when analyzing customer reviews, social media conversations or customer feedback.
 - ✓ E.g. if the words **expensive, overpriced and overrated** frequently appear on your customer reviews, it may indicate you need to adjust your prices (or your target market!).
- **Collocation:** refers to a sequence of words that commonly appear near each other.
 - ✓ Types of collocations are **bigrams** (a pair of words, like get started, save time or decision making) and **trigrams** (a combination of three words, like within walking distance or keep in touch).

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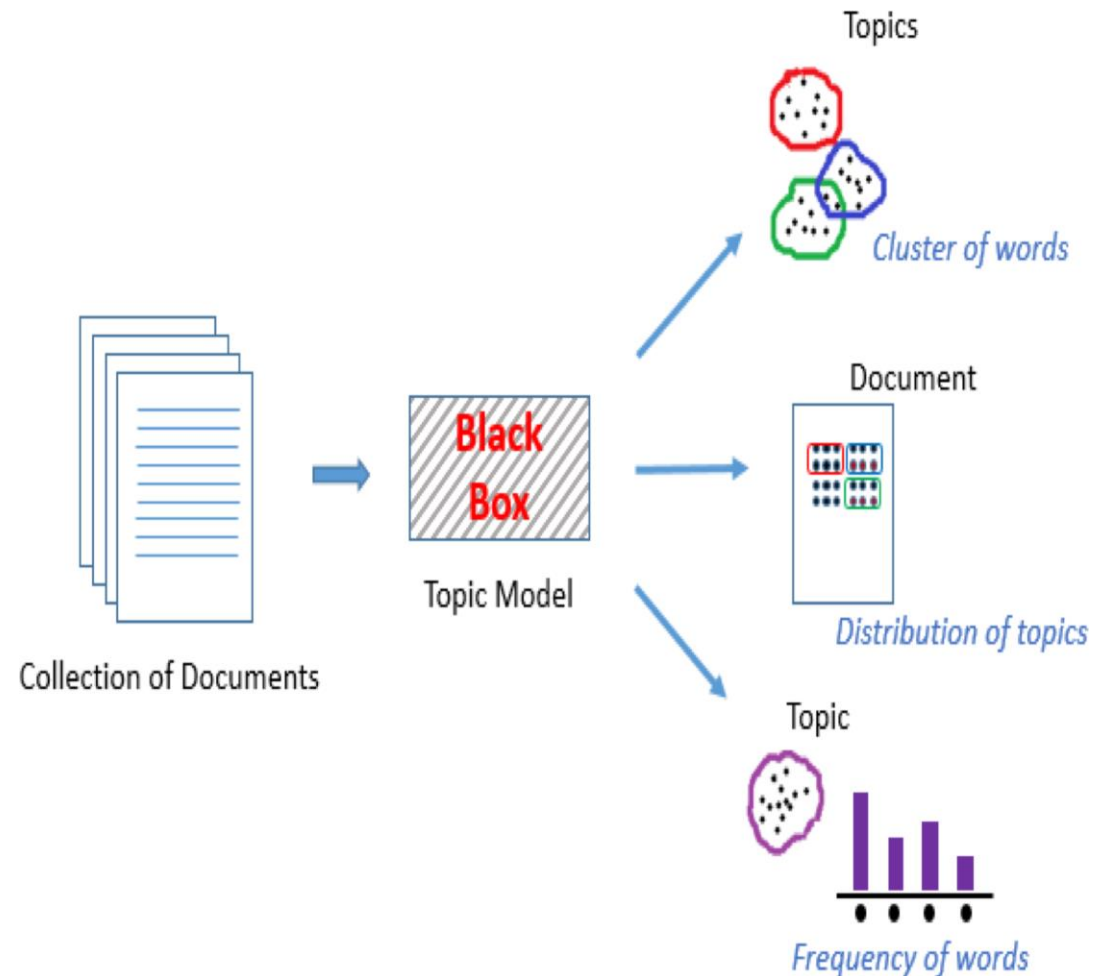
- ✓ Identifying collocations — and counting them as one single word — improves the granularity of the text, allows a better understanding of its semantic structure and, in the end, leads to more accurate text mining results.
- ❖ **Concordance**: used to recognize the particular context or instance in which a word or set of words appears.
 - ✓ The same word can be used in different contexts and concordance allow to understand its exact meaning based on context.
- ❖ **Text Classification**: process of assigning categories (tags) to unstructured text data.
- ❖ **Text extraction**: process of extracting specific pieces of data from a text, like keywords, entity names, addresses, emails.

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- The most popular tasks of text classification – topic analysis, sentiment analysis, language detection, and intent detection.
- The main tasks of text extraction – keyword extraction, named entity recognition and feature extraction.

Topic Analysis/Modeling

- A technique to discover topics across various text documents.
- Helps to understand the main themes or subjects of a text.
- Is one of the main ways of organizing text data (**Digital Library**).
- Words which are related to each other form a topic.
- Similarly, there can be **multiple** topics in an individual document.



Sentiment Analysis

- The process of computationally identifying and categorizing opinions, expressed in a text, towards a particular topic, product, place or person.
- Helps to understand the opinion and feelings in a text, and classify them as positive, negative or neutral.
- Analyze and classify customer conversation based on:
 - ✓ Key aspects of a product and service that customers care about.
 - ✓ Users' underlying intentions and reactions concerning those aspects.

Language Detection

- Allows to classify a text based on its language.
- One of its most useful applications is automatically routing support requests to the right geographically located team.

Keyword Extraction

- Keywords are the most relevant terms within a text and can be used to summarize its content.
- Keyword extractor allows you to index data to be searched, summarize the content of a text or create tag clouds, among other things.

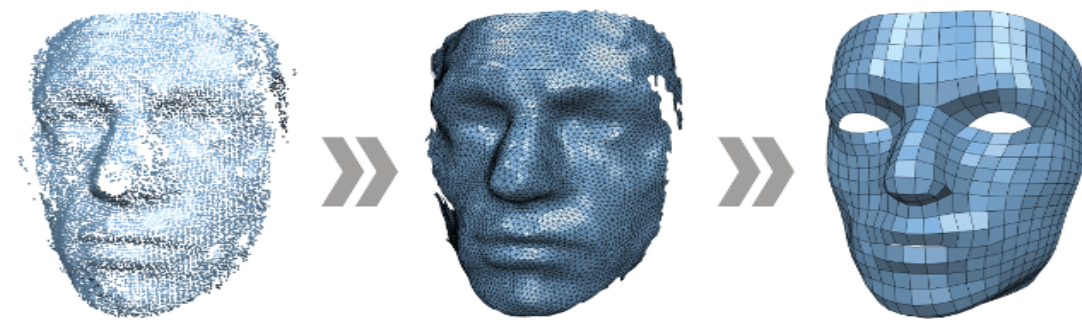
Named Entity Recognition

- Locate and classify named entities in text into pre-defined categories such as the names of persons, organizations, locations, expressions of times, quantities, monetary values, percentages, etc.
- Allows to identify and extract the names of companies, organizations or persons from a text.
- Help answering many real-world questions, such as:
 - Which companies were mentioned in the news article?
 - Were specified products mentioned in complaints or reviews?
 - Does the tweet contain the name of a person?
 - Does the tweet contain this person's location?
- It can be implemented as text preprocessing.

Feature Extraction

- Helps identify specific characteristics of a product or service in a set of data.
- For example, if you are analyzing product descriptions, you could easily extract features like color, brand, model, etc.

Image Analytics



- Image analysis deals with the extraction of image patterns from a large collection of images.
- There are two types of methods used for image processing namely, **analogue** and **digital** image processing.
- Analogue image processing can be used for the hard copies like printouts and photographs.
- Digital image processing techniques help in manipulation of the digital images by using computers.

Application of Image Analysis

- Image enhancement
- Image restoration
- Image compression
- Image Segmentation
- Image Detection/Object recognition
- Image Classification

Image enhancement



- Used to refine a given image so that desired image features become easier to perceive for the human visual system or to be detected by automated image analysis systems.
- Image enhancement refers to techniques that improve the visual quality of an image, making it more suitable for analysis or presentation.
- Allows the observer to see details in images that may not be immediately observable in the original image.
- is the transformation or mapping of one image to another.
- Valuable image information may be lost, or the enhanced image may be a poor representation of the original.

Image restoration

- Defined by **estimating** an original image from data that has been degraded in some way.
- Image restoration aims to **recover** an image that has been degraded by factors like **noise**, **blur**, or **loss of resolution**.
- Refers to the recovery of an image from its degraded version.
- Depending on the degradation model, image restoration includes inpainting, deblurring, denoising, and so on.



Image compression



- Is the application of data compression on digital images.
- The objective is to **reduce** redundancy of the image data to be able to store or transmit data in an efficient form.
- Image compression reduces the size of an image file without significantly degrading quality, making storage and transmission more efficient.
- Is useful for image analysis, image storage, and image transmission.

Image Segmentation

Object
Detection



Instance
Segmentation



- Image segmentation involves **partitioning** an image into multiple segments or regions to locate objects and boundaries in images to simplify analysis.
- The prior goal of segmentation is to make things simpler and transform the representation of images into a meaningful subject.
- For example, before a tumor is analyzed in a computed tomography (CT) volumetric image, it has to be detected and somehow isolated from the rest of the image.
- Before a face is recognized, it has also to be picked out from its background.

Object Recognition/Detection

- Locate the presence of objects in an image.
- Produce a list of object categories present in the image along with an axis-aligned bounding box indicating the position and scale of every instance of each object category.

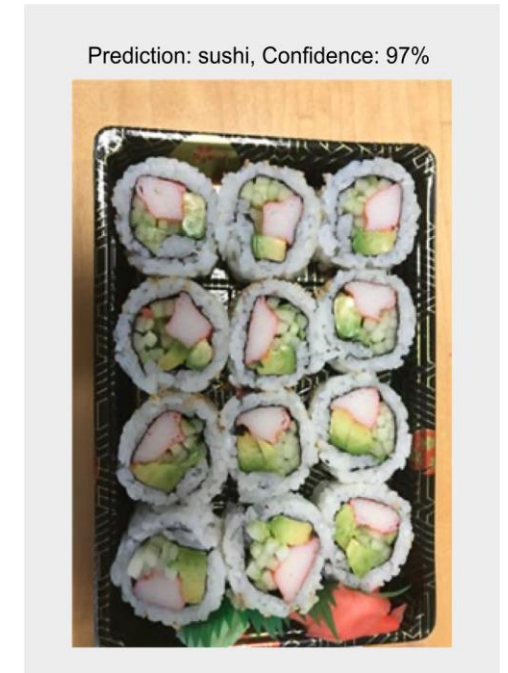
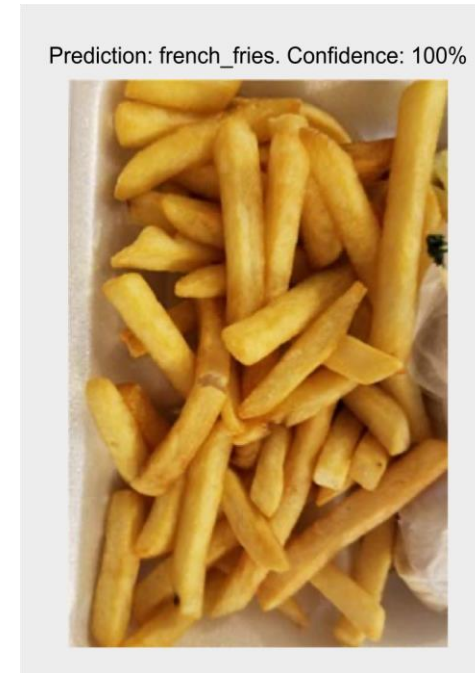
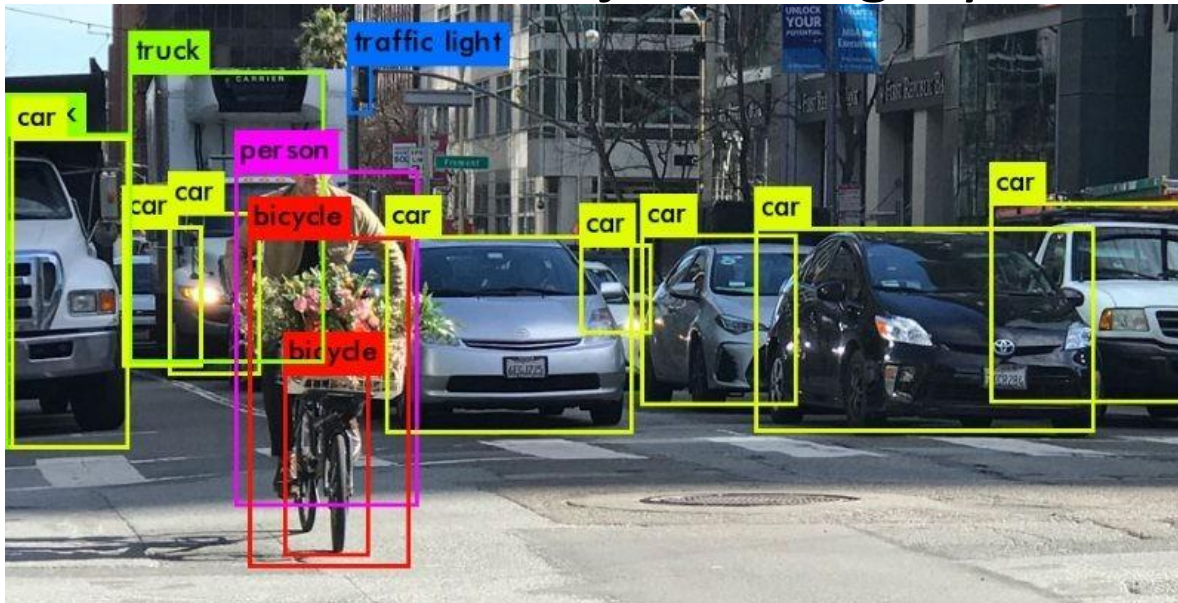
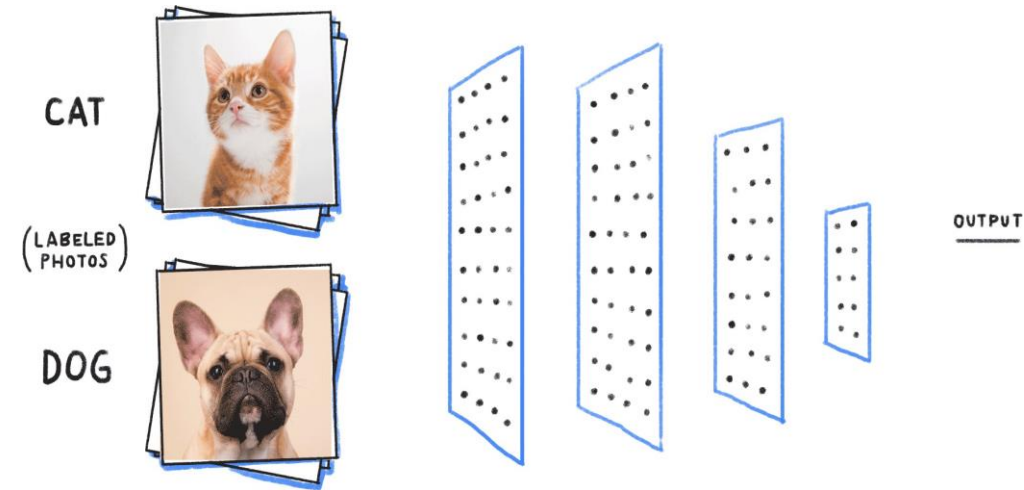


Image Classification

- Refers to a process of classify an image according to its visual content.
- E.g., tell if an image contains a cat or dog.
- The classification can be binary, when the algorithms have to decide if an image contains a tumor/not.
- Otherwise, the classification can be multiclass when the algorithms have to grade the pathologic stage of cancer.



Stream Analytics/Processing



- We have entered an era where competitive advantage comes from analyzing, understanding, and responding to an organization's data.
- Time is of the essence, and speed will decide the winners and losers.
- The reality is that the **value of most data degrades with time**.
- It's interesting to know that yesterday there was a traffic jam, or 10 fraud incidents, or 10 people who had heart attacks.
- However, it is much better if we can gain those insights **at the time they are occurring** so that we can intervene and manage the situation.

Cont'd



- Streaming Data:
 - is data that is **generated continuously** by thousands of data sources.
 - Is the processing of **data in motion**.
- Stream processing is **computing on data directly as it is produced or received**.
- is used to **query continuous data stream** and **detect conditions within a small time period**.
- The majority of data are born as continuous streams over time.

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- Streaming data includes a wide variety of data such as:
 - ✓ Log files generated by customers using mobile or web applications.
 - ✓ E-commerce purchases.
 - ✓ In-game player activity.
 - ✓ Information from social networks.
 - ✓ Financial trading floors.
 - ✓ Geospatial services.
 - ✓ Telemetry from connected devices.

Cont'd

- The application logic, analytics, and queries exist **continuously**, and data flows through them continuously.
- Upon receiving an event from the stream, a stream processing application reacts to that event:
 - ✓ It may trigger an action, update an aggregate or other statistic, or “remember” that event for future reference.



Cont'd

- Stream analysis can also **process multiple data streams jointly**, and each computation over the event data stream may produce other event data streams.
- The systems that receive and send the data streams and execute the application or analytics logic are called ***stream processors***.
- The basic responsibilities of a stream processor are to ensure that **data flows efficiently** and the **computation scales** and is **fault tolerant**.

Cont'd

- Stream processing unifies Data Collection, Processing, Analytics, and Applications:
 - ✓ Classifying a banking transaction as fraudulent based on an analytical model *then* automatically blocking the transaction.
 - ✓ Sending push notifications to users based on models about their behavior.
 - ✓ Adjusting the parameters of a machine based on result of real-time analysis of its sensor data.

Application of Stream Processing

- Alerts and Thresholds-detects a condition (simple or complex) and generates alerts based on a condition (e.g. Alarm on high temperature).
- Detect the rise or fall, outliers of a value of stock markets.
- Simple Counting and Counting with Windows (aggregate functions)- Detect a peak or a trough.
- Detect chart patterns for stock markets, patient monitoring.
- Surveillance-Identify a behavior signature (e.g. walking, running).
- Tracking-fleet of vehicles, wildlife, airline luggage.

Cont'd

- Joining Event Streams-football game with both the players and the ball having sensors that emit events with current location and acceleration.
- Data Correlation, Missing Events, and Erroneous Data:
 - ✓ Matching up two data streams that send events at different speeds
 - ✓ Detecting a missing event in a data stream (e.g. detect a customer request that has not been responded within 1 hour of its reception, check for the completeness of a task. (Does the reply to an email come within an hour?))
 - ✓ Detecting erroneous data (e.g. Detect failed sensors using a set of sensors that monitor overlapping regions and using those redundant data to find erroneous sensors and removing their data from further processing).
- Context-aware promotions and advertising.

Batch Vs Stream Processing

	Batch processing	Stream processing
Data scope	Queries or processing over all or most of the data in the dataset.	Queries or processing over data within a rolling time window, or on just the most recent data record.
Data size	Large batches of data.	Individual records or micro batches consisting of a few records.
Performance	Latencies in minutes to hours.	Requires latency in the order of seconds or milliseconds.
Analyses	Complex analytics.	Simple response functions, aggregates, and rolling metrics.

Challenges Addressed by Stream Analysis

- **Applications and analytics react to events instantly:**
 - ✓ There's no lag time between "event happens" -> "insight derived" -> "action is taken".
 - ✓ Actions and analytics are up-to-date, reflecting the data when it is still fresh, meaningful, and valuable.
- **Stream processing can handle data volumes that are much larger than other data processing systems:**
 - ✓ Some data naturally comes as a never-ending stream of events, not even possible to store it.
 - ✓ Batch processing lets the data build up and try to process them at once.
 - ✓ The event streams are processed directly, and only a meaningful subset from the data is persisted.

Cont'd

- **Stream processing naturally and easily models the continuous and timely nature of most data:**
 - ✓ Stream processing naturally fit with time series data and detecting patterns over time.
 - ✓ Incrementally computing updates, rather than periodic re-computation of all data fits naturally with the stream processing model.
- **Stream processing decentralizes and decouples the infrastructure:**
 - ✓ The streaming paradigm reduces the need for large and expensive shared databases.
 - ✓ Instead, each stream processing application maintains its own data and state, which is made simple by the stream processing framework.
 - ✓ In this way, a stream processing application fits naturally in a microservices architecture.

Assignment1

1. How can machine learning techniques be applied to improve image analytics tasks such as detection, segmentation, and classification? Discuss specific algorithms that are commonly used?
2. How does Natural Language Processing (NLP) facilitate the transformation of unstructured text into actionable information?
3. Explain the Strength and weaknesses of Stream Processing with example.